

Faculty Profile

Name: **Dr. SACHIN DHULL**

Designation: Assistant Professor

Qualification: Ph.D, M.Tech, B.Tech (Mechanical)

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Area of Interest/Specialization: Abrasive Flow Machining, Computer-Integrated Manufacturing, Computer-Aided Design, Computational Fluid Dynamics, Finite Element Modelling, Industrial Internet of Things, Data Science, Machine Learning and Artificial Intelligence

Experience: 11.5 years

Academic qualification details:

S.No.	Qualification	Specialization	CGPA/%	Institution	Passing year
1.	Ph.D	Advanced Manufacturing	8 CGPA	Delhi Technological University	2021
2.	M.Tech	Manufacturing	8.3 CGPA	YMCAUST Faridabad	2014
3.	B.Tech	Mechanical	69%	Kurukshetra University	2012
4.	12 th	Non-Medical	78.4%	AKSIP School, Chandigarh (CBSE)	2006
5.	10 th		83%	AKSIP School, Chandigarh (CBSE)	2004

Experience details:

S.No.	Post held	Institution	Period (from)	(till)	Duration
1.	Assistant Professor	Maharaja Surajmal Institute of Technology	01-01-2021	Till date	5 years, 8 months
			13-01-2020	25-04-2020	3 months, 12 days
			01-08-2019	20-12-2019	4 months, 20 days
			30-07-	31-05-	10 months

			2018	2019	
2.	Assistant Professor	Delhi Technical Campus, Bahadurgarh	17-08-2017	28-07-2018	1 year
3.	Assistant Professor	Lingaya's University, Faridabad	01-08-2017	10-08-2017	10 days
4.	Teaching-cum Research Associate	Delhi Technological University	09-07-2014	31-07-2017	3 years
5.	Lecturer	Government Polytechnic College, Jhajjar	Feb., 2012	May, 2012	4 months
6.	Trainee	Zenner Aquamet India Pvt. Ltd.	08-08-2011	15-07-2012	1 year
Total experience:					11.5 years

Coaching experience details:

S.No.	Post held	Institution	Period (from)	(till)	Duration
1.	Faculty for English	Success Heights Academy for Competitive Exams	01-05-2019	31-07-2019	3 months
2.	Faculty for GATE ME	GATE Coaching Institute, Bahadurgarh	July, 2017	Dec., 2017	6 months
3.	Faculty for GATE ME	Dronacharya Institute of Engineers for GATE, Katwaria Sarai, Delhi	Jan., 2016	June, 2016	6 months
Total experience:					1 year, 3 months

GATE Exam details:

S.No.	GATE	Year	Marks (out of 100)
1.	XE	2025	39.67
2.	ME	2023	29.33
3.		2022	30.47
4.		2019	42.08
5.		2018	35.52
6.		2017	48.71
7.		2016	48.78
8.		2015	39.31
9.		2014	44.72
10.	PI	2011	38.67

Management entrance competition exam details:

S.No.	Exam	Year	Percentile (Quantitative ability section)
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1.	CMAT	2016	91.03
2.	CAT	2015	80.42

Other competitive exam detail:

S.No.	Exam	Year	Marks (out of 150)
1.	HTET (Level-2 TGT)	2019	91

Key Publications:

Publications in SCI indexed Journals:

1. Maurya, N. K., Saxena, A., Dwivedi, S. P., Beemkumar, N., **Dhull, S.**, Sharma, S., ... & Gupta, D. (2025). Experimental investigation on implications of SLM-based 3D printing parameters on the grain morphology, phase formation, fracture deformation, and mechanical properties of 316 L stainless steel: finite element analysis and parametric optimizations. *Applied Physics A*, 131(8), 639.
2. Maurya, N. K., **Dhull, S.**, Maurya, M., & Vishnoi, M. (2026). Optimization of Tensile and Flexural Performance of Carbon Fiber-Reinforced Polyethylene Terephthalate Glycol Fabricated via Fused Deposition Modeling. *Journal of Materials Engineering and Performance*, 1-10.
3. Chaudhary, S., Singh, S., Gupta, R., & **Dhull, S.** (2026). Role of Artificial Intelligence in Bioinformatics: Revolutionizing Molecular Docking and DNA Tokenization. *Computational Biology and Chemistry*, 109217.
4. **Sachin Dhull**, RS Walia, Qasim Murtaza, MS Niranjana, Experimental, computational and mathematical analysis of hybrid abrasive flow machining process, International Journal of Precision Engineering and Manufacturing, Springer publishing, 22(10), pp. 1657-1680, 2021.
5. A Bhardwaj, **Sachin Dhull**, S Kumawat, GP Joshi, D Gupta, D Yadav, Analysis of aluminium-sic hybrid composite based on banana stem fibres, Green Materials, Emerald publishing limited, pp.1-19, 2025
6. U Punia, **Sachin Dhull**, A Habeeb, E Antil, D Dahiya, D Gupta, A Ahlawat, Adopting additive manufacturing: spotlight on sustainability and material utilization, Green Materials, Emerald publishing, pp.1-11, 2025
7. NK Maurya, M Maurya, M Vishnoi, V Singh, A Bansal, **Sachin Dhull**, A saxena, Experimental investigation and optimization of process parameters on tensile strength of carbon coated PLA material, Physica Scripta, IOP publishing, 99(12), 2024
8. A Tyagi, **Sachin Dhull**, R Nair, GP Joshi, BL Gupta, SL Meena, Assessment of surface texturing for sliding wear and corrosion resistance on sustainable HVOF sprayed Al₂O₃+C powder coating, Emerging Materials Research, Emerald publishing, 2025
9. **S Dhull**, S Kumar, AP Singh, V Kumar, MS Pawar, Modelling and optimization of electrical discharge machining parameters for titanium alloy using copper electrode, International Journal of Maritime Engineering, 2026

10. **S Dhull**, S Kumar, J Raj, R Mishra, N Kulkarni, Tribological and microstructural characterization of LPBF fabricated AlSi10Mg composites with BC reinforcement, International Journal of Maritime Engineering, 2026
11. **S Dhull**, AP Singh, VC Dinkar, BK Nagappan, I Molvi, Synthesis, mechanical properties and microstructural analysis of Al2024/ SiC/Al₂O₃/Sn hybrid metal matrix composites, International Journal of Maritime Engineering, 2026

Publications in ESCI indexed Journals:

12. **Sachin Dhull**, RS Walia, Qasim Murtaza, MS Niranjana, Electrochemo magneto abrasive flow machine setup fabrication and experimental investigation of the process alongwith mathematical modeling and optimization, Independent journal of management and production, 10(8), pp.1834-1849, 2019
13. Malvika Chaudhary, **Sachin Dhull**, Sobinder Singh, Rekha Tripathi, Comprehensive review of nanomaterial methods, characterization and multifaceted applications in medical field, Journal of polymer and composites, 2025
14. Tripathi, R., Chaudhary, M., Singh, S., & **Dhull, S.** (2025). Phytochemical and Antimicrobial Screening of Stem Bark of Thevetia peruviana. *Oriental Journal of Chemistry*, 41(5).

Publications in Scopus indexed Journals:

15. S Singh, U Singh, N Singh, A Baliyan, **S Dhull**, H Tomar, Neetu, Advanced predictive analytics for used car pricing through convolutional neural networks, International Journal of environmental sciences, 11(22), 2025.
16. **Sachin Dhull**, RS Walia, Qasim Murtaza, MS Niranjana, Development of different polymer media and experimental investigation of abrasive flow machining process, International Journal of engineering research and technology, 11(7), pp.1015-1024, 2018
17. **Sachin Dhull**, Sandeep Singh, Static hand gesture based human computer interaction using convolution neural network, International Journal of research, 10(12), 2023

Publications in other peer-reviewed Journals:

18. Rajbir Singh, **Sachin Dhull**, White topping overlay for rehabilitation of bituminous roads, International Journal of research, 10(11), 2023
19. Vikash, **Sachin Dhull**, Experimental investigation of the effect of welding parameters on mild steel specimens in MIG welding process alongwith metallurgical and mechanical analysis, International Journal of research, 9(11), 2022
20. Rajbir Singh, **Sachin Dhull**, Causes and repair techniques for distresses in concrete pavements, International Journal of research, 9(6), 2022
21. Rajbir Singh, **Sachin Dhull**, Study of hybrid abrasive flow machining process alongwith non traditional machining processes: a review, International Journal of research, 2021

22. Jyoti Vashishta, **Sachin Dhull**, Experimental determination of compressive, split tensile and flexural strength of HFRC using steel and propylene fibres in different proportions, Journal of engineering research and applications, 8, pp.44-51, 2018
23. Rajbir Singh, **Sachin Dhull**, Magnetic abrasive flow machining process review and its experimental investigation, International journal of engineering research and technology, 2019
24. **Sachin Dhull**, RS Walia, Harwinder Singh, Experimental study of process parameters for magnetic assisted AFM incorporating the drill bit helical effect, International Journal of research, 2016
25. **Sachin Dhull**, RS Walia, Study of magnetic assisted AFM, mechanical properties of various abrasive laden polymer media and abrasive wear and force mechanism, International journal of advance research and innovation, 2016
26. Arvind, Rajkumar Duhan, **Sachin Dhull**, Development and parametric optimization of abrasive flow finishing (AFF) process of aluminium workpiece using nylon fixture, International journal of all research education and scientific methods, 2016
27. **Sachin Dhull**, RS Walia, Qasim Murtaza, MS Niranjana, Modeling and simulation of AFM experimentation of brass workpiece using different media, International journal of latest engineering and management research, 2019
28. **Sachin Dhull**, RS Walia, Qasim Murtaza, MS Niranjana, Experimental investigation of abrasive flow machining process alongwith fuzzy logic and grey relation analysis, International journal of latest engineering and management research, 2019
29. **Sachin Dhull**, RS Walia, Qasim Murtaza, MS Niranjana, Experimental investigation and preparation of electromagnets and their effect on surface roughness in abrasive flow machining process, International journal of latest engineering and management research, 2019

Papers presented in Conferences:

S.No.	Paper title	Conference details	Date	Place
1.	Mechanical aspects of artificial intelligence algorithms and mathematical and experimental investigation of hybrid abrasive flow machining process	3 rd International Conference on Artificial Intelligence and Applications	26-27 March, 2025	Maharaja Surajmal Institute of Technology
2.	A comprehensive review on climate change mitigation strategies and their impact on achieving sustainability goals	International Conference of Advance Research and Innovation	Feb., 2025	Delhi State Centre, Institution of Engineers (India), New Delhi
3.	From tradition to innovation: Artificial intelligence and machine learning in molecular docking	National Conference on Recent Trends in Climate Change, Energy and Sustainability	Nov., 2014	Maharaja Surajmal Institute of Technology

4.	Ecological conservation, biodiversity, environmental laws, pollution and environmental impact assessment in India	National Conference on Recent Trends in Climate Change, Energy and Sustainability	Nov., 2024	Maharaja Surajmal Institute of Technology
5.	Enhancing educational outcomes: Student performance analysis using ETL process	International Conference on technological innovative transformation to industry 5.0	May, 2024	Dronacharya College of Engineering, Gurugram
6.	Abrasive flow machining process hybridization with other non-traditional machining processes: a review	International Conference on mechanical and energy technologies	Nov., 2019	Galgotias College of Engineering and Technology, Greater Noida
7.	Modelling of slag produced in submerged arc welding	International Conference on emerging trends in electro-chemical technologies and management	July, 2019	HMRITM, New Delhi
8.	Development, characterization and optimization of newly developed polymer media and its computational flow dynamics of abrasive flow machining	3 rd International Conference on advanced production and industrial engineering	Oct., 2018	Delhi Technological University
9.	Hybrid abrasive flow machining process models and their computer and industrial applications	International Conference on advanced production and industrial engineering	Dec., 2016	Delhi Technological University
10.	Synthesis CNTs particle based abrasive media for abrasive flow machining process	National Conference on processing and characterization of materials	Dec., 2015	NIT Rourkela

FDP attended:

S.No.	Programme	Topic	Organizing institute	Date	duration
1.	AICTE FDP	Effective teaching-learning using social media	NITTTR Chandigarh	May, 2025	1 week

2.	AICTE FDP	Online learning and evaluation	NITTTR Chandigarh	Dec., 2024	1 week
3.	AICTE FDP	Energy storage materials and technology	NITTTR Chandigarh	May, 2024	1 week
4.	AICTE FDP	Stress management	NITTTR Chandigarh	Jan., 2024	1 week
5.	Training programme	Performance optimization of RIS assisted communication network using reinforcement learning algorithms for smart villages	Delhi Technological University	July, 2023	1 day
6.	Memorial seminar	Birth anniversary of Dr. Jag Jeet Singh Rathi	Maharaja Surajmal Institute of Technology	May, 2023	1 day
7.	Training programme	IP awareness	Intellectual Property Office	March, 2022	1 day
8.	Hybrid mode workshop	Standardization of healthcare and medical devices	CSIR-NPL	Dec., 2021	2 days
9.	AICTE Training and Learning (ATAL) Academy	Smart manufacturing through digital manufacturing	Centurion University of Technology and Management	Nov., 2021	1 week
10.	TEQIP III FDP	Digital manufacturing	Punjab Engineering College, Chandigarh	Dec., 2020	1 week
11.	AICTE FDP	Planning and management for institutional excellence	NITTTR Chandigarh	Nov., 2019	1 week
12.	FDP	Contemporary issues and challenges in management	Maharaja Surajmal Institute of Technology		
13.	TEQIP III FDP	International lecture series on bondgraph modelling and simulation	Delhi Technological University	Oct., 2018	1 week
14.	Workshop	Solidworks	SAE-DTU	Sept., 2017	2 days
15.	TEQIP II FDP	Recent development and challenges in materials and manufacturing processes	Delhi Technological University	July, 2016	1 week
16.	National seminar	Innovations and challenges in thermal engineering	Delhi Technological University	July, 2016	2 days
17.	TEQIP II short term training	PLC, HMI, SCADA & AC Drives	Delhi Technological	June, 2016	1 week

	program		University		
18.	TEQIP FDP	Quality assurance in engineering and technology education	DCRUST, Murthal	Dec., 2015	1 week
19.	TEQIP II FDP	Automation in manufacturing	Delhi Technological University	May, 2015	2 weeks
20.	TEQIP II FDP	Precision manufacturing: technology for better tomorrow	Delhi Technological University	July, 2014	1 week

Courses/training:

S.No.	Training	Organized by	From	to	Duration
1.	Faculty Induction Programme	Ramanujan College, University of Delhi	21-02-2024	19-03-2024	4 weeks
2.	Vocation Training	Tractor Division, HMT Pinjore	06-07-2009	14-08-2009	6 weeks
3.	Industrial Training (AUTOCAD)	SQL Technologies	July, 2009	Aug., 2009	4 weeks

Organizing committee member in conferences:

S.No.	Responsibility held	Conference	Duration	Place
1.	Organizing Committee member	3 rd International Conference on Artificial Intelligence and Applications	26-27 March, 2025	Maharaja Surajmal Institute of Technology
2.	Organizing Committee member	2 nd International Conference on Artificial Intelligence and Applications	19-20 March, 2024	Maharaja Surajmal Institute of Technology
3.	Organizing Committee member	7 th National Conference on Advances in Metrology	5-6 March, 2021	Maharaja Surajmal Institute of Technology
4.	Volunteer	International Conference on Artificial Intelligence and Applications	6-7 Feb., 2020	Maharaja Surajmal Institute of Technology
5.	Organizer	International Conference on Recent Advances in Mechanical Engineering	14-15 Oct., 2016	Delhi Technological University

University level Duties:

S.No.	Duty	University	Centre
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1.	Centre Suprintendent	M.D.U. B.Tech Exam, May-June, 2018	Akido College of Engineering, Bahadurgarh
2.	University Representative	GGSIU	
3.	Evaluator	GGSIU	
4.	External Examiner	GGSIU	

Institute/Departmental level duties:

S.No.	Duty	S.No.	Duty	S.No.	Duty
1.	Mentor	7.	Admission	13.	Media cell
2.	Proctor	8.	NBA	14.	Print media
3.	Result analysis	9.	Communication lab set-up	15.	DTC bus pass
4.	Furniture and computer record	10.	Purchase committee	16.	e-waste
5.	Time table committee	11.	Refreshment committee	17.	Anti-ragging
6.	Discipline	12.	Music-system		

Other details:

- Secured AIR 192 in GATE 2011 and got admission seat in IIT Roorkee
- Published 2 books
- Published 2 patents
- Guided 1 M.Tech student
- Guiding 1 Ph.D student
- Got DTU research excellence award 2021

Blood donation details:

S.No.	Institute	Place	Date
1.	Ashirwad Blood Center	Bahadurgarh	Aug., 2025
2.	Pitampura blood centre	Maharaja Surajmal Institute	April, 2022
3.	Rotary blood bank	Maharaja Surajmal Institute	Feb., 2019
4.	PGIMR Chandigarh	SDDIET Barwala	Nov., 2007

Participation in school level competitions:

S.No.	Competition	Organizing Committee	Date
1.	National Group Song Competition (Team was adjudged second)	Parishad, Shivalik Public School, Chandigarh	29-09-2004
2.	Inter-house 4×100 mts. Relay race (boys) (3 rd position)	Ajit Karam Singh International Public School, Chandigarh	09-01-2003

3.	4 th National Science Olympiad	Science Olympiad Foundation	01-02-2002
4.	Lay's Just A Minute Extempore Speaking Competition	Chandigarh	21-07-2001
5.	English speaking & personality development workshop	Chandigarh	09-07-2001
6.	On-the Spot Inter-School Essay Writing Competition	Generation Saviour Association, Mohali	
7.	On-the Spot Embroidery Contest	Anchor Educational Service, Chandigarh	30-04-2001
8.	Hindi Essay Competition	Aeronautical Society of India, Chandigarh	April, 2001
9.	Intra-School Bournvita Quiz Contest	Cadbury India Limited, Chandigarh	April, 2001
10.	3 rd National Science Olympiad	Science Olympiad Foundation	29-01-2001
11.	On-the Spot Painting Competition	The Times of India Luxor Colour Splash, Rock Garden, Chandigarh	24-01-2001
12.	Intra-School Bournvita Quiz Contest	Cadbury India Limited, Chandigarh	19-10-2000
13.	Hindi Essay Writing Competition	Shrimati Kamla Ghai Smriti Trust, Arya Samaj, Chandigarh	20-08-2000
14.	Lay's Just A Minute Extempore Speaking Competition	Chandigarh	28-02-2000
15.	Students' Talent Promotion Examination	Children's Welfare Trust of India, Chandigarh	Jan., 2000
16.	Dainik Bhaskar Cheers Chandigarh Painting/Essay Competition	Chandigarh	Dec., 1999
17.	On-the Spot Inter-School Essay Writing Competition	Ydavindra Public School, Mohali	May, 1999

Personal details:

Name: Sachin Dhull

D.O.B.: 28th Nov., 1989

Extra-curricular activities: singing, playing cricket